SEQUENCE 9

# Proportions and percentage change

21–29 January 2027

## 9.1 Proportion and proportion of a proportion

**Key point · Definitions.** In a nonempty finite set $E$, the proportion of a subset $F$ is $p = \text{Card}(F) / \text{Card}(E)$; $0 \leq p \leq 1$. Its expression as a percentage is $100p\,\%$. If $G \subseteq F \subseteq E$, $F \neq \varnothing$, and if $p_1$ is the proportion of $F$ in $E$ and $p_2$ that of $G$ in $F$, the proportion of $G$ in $E$ is $p_1 p_2$.

**Example 1.** A school has 1500 students, of whom 652 are day boarders. Calculate their proportion. In a class, 60 % of the students play sport; among them, 40 % are boys. What proportion of the class do the boys who play sport represent?

**Example 2.** In a fictional school of 800 students, 35 % play a sport at a club; among them, 45 % play a team sport. Calculate the two counts and the final proportion.

**Watch out.** Proportion and rate of change do not answer the same question: a proportion compares a part to a whole, a rate compares a change to an initial value.

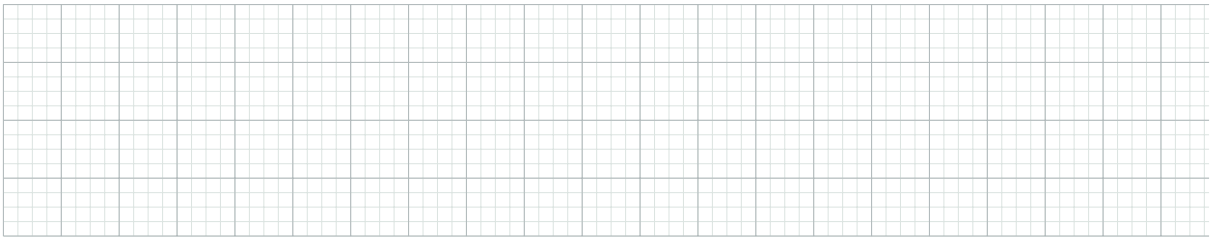
## 9.2 Absolute change and rate of change

**Key point · Definition.** For an initial value $V_I > 0$ and a final value $V_F > 0$, the absolute change is $V_F - V_I$; the rate of change is

$$t = \frac{V_F - V_I}{V_I}.$$

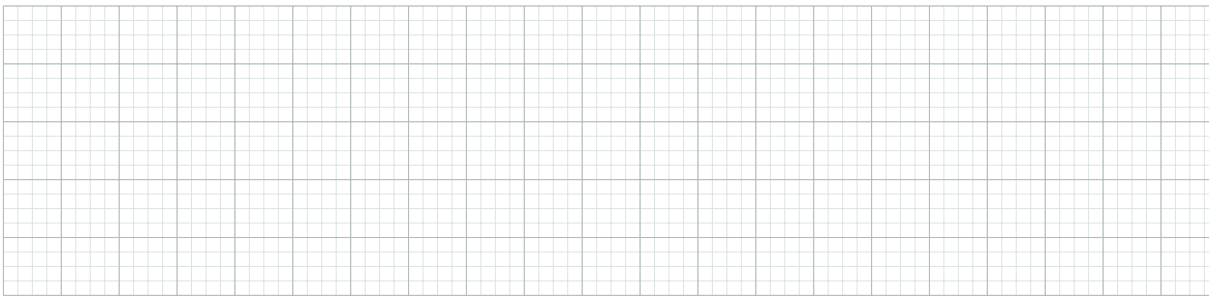
The rate has no unit. It is positive for an increase, negative for a decrease. The reference is always the initial value.

**Example 3.** A cheese is sold for 2.13 euros instead of 3.55 euros. Calculate the absolute change and the rate.



**Key point · Multiplier.** $CM = V_F/V_I = 1 + t$, so $V_F = V_I(1 + t)$. To recover $V_I$, divide $V_F$ by $1 + t$ (nonzero).

**Example 4.** A fictional item priced 50000 Ar increases by 12 %, then separately consider a 12 % discount on the initial price. Give the final prices. A price after a 20 % increase is 72000 Ar; find its initial price.



**Fluency · 5–10 min.** Expand $(x - 3)^2$; solve $1/x = 5$; give the mean of $8; 10; 12$.

## 9.3 Successive changes

**Key point · Property.** Successive changes with rates $t_1, t_2$ have overall multiplier

$$CM_G = (1+t_1)(1+t_2), \qquad t_G = (1+t_1)(1+t_2) - 1.$$

We multiply the multipliers; we do not simply add the rates.

**Example 5.** A manufacturer sold 5600 bicycles in 2022, which is 25 % more than in 2021. Its sales decrease by 15 % in 2023. Determine the sales of 2021 and 2023, then the overall rate between these two years.

**Example 6.** A price increases by 10 % then decreases by 10 %. Does it return to the initial price?

**Example 7.** A rate goes from 20 % to 25 %. Explain the difference between the increase in percentage points and the relative change.

## 9.4 Reverse change

**Key point · Property.** To return to the initial value after a change of rate $t > -1$, use the multiplier $1/(1+t)$. The reverse rate is

$$t_{\text{reverse}} = \frac{1}{1+t} - 1 = \frac{-t}{1+t}.$$

**Example 8.** Which rate compensates a 25 % increase? Which rate compensates a 20 % decrease?

**Example 9.** A price undergoes a 30 % discount, then a 10 % increase. What additional percentage increase restores the initial price?